

# Intrabeam Collisions, Gas and Electron Effects

USPAS - Beam Physics with Intense Space Charge

Austin Hoover   Steven Lund   Daniel Winklehner   John Barnard

July 2026

# Outline I

1. Coulomb collisions
2. Beam-gas scattering
3. Charge-changing processes
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability

# Outline

1. Coulomb collisions
2. Beam-gas scattering
3. Charge-changing processes
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability

# 1.1 Gas and electron effects

- Effects depend strongly on particle mass and charge.
- Timescales are much different in circular accelerators vs. linacs ( $t_{residence} \sim$  milliseconds to days vs. 10's of microseconds).
- Long vs. short pulse ( $t_{pulse} \sim$  10's of microseconds vs. 10's of nanoseconds).

# 1.2

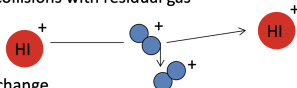
|                                                                                             |                                                                                                         |                |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------|
|  Heavy ion |  Residual gas molecule | $e^-$ electron |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------|

Processes:

1. Coulomb collisions (intra-beam)



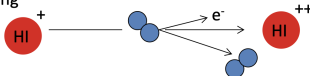
2. Coulomb collisions with residual gas



3. Charge exchange



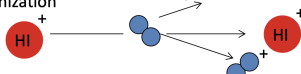
4. Stripping



5. Neutralization

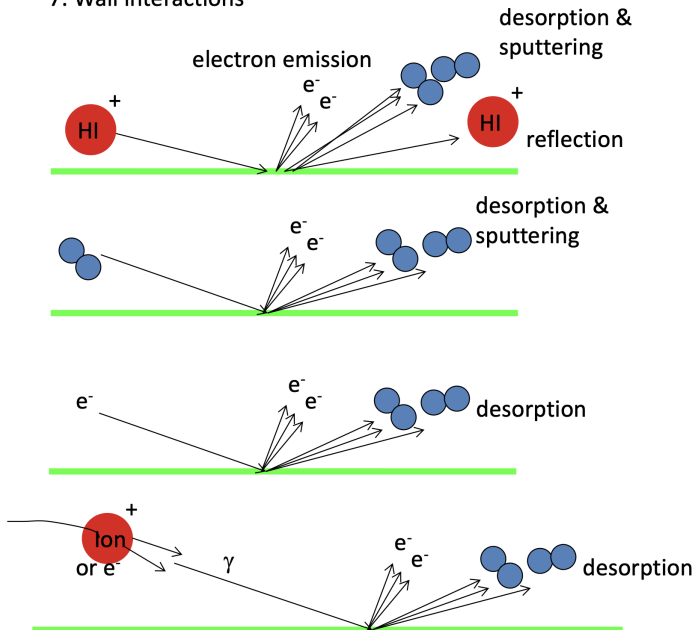


6. Gas Ionization



# 1.3

## 7. Wall interactions



## 1.4 Coulomb collisions

Consider effects of Coulomb collisions in a continuous beam propagating through a continuous focusing channel with  $T_{\perp} \neq T_{\parallel}$ . (If  $T_{\perp} \neq T_{\parallel}$ , the beam is already relaxed.)

From Ichimaru and Rosenbluth [1]:

$$\frac{dT_{\perp}}{dt} = -\frac{1}{2} \frac{dT_{\parallel}}{dt} = \frac{T_{\parallel} - T_{\perp}}{\tau}.$$

(Since  $T_x = T_y = T_{\perp}$ ,  $T_{\parallel}$  changes at twice the rate of  $T_{\perp}$ . And since  $2k_B T_{\perp} + k_B T_{\parallel} = \text{constant}$ .)

$\tau$  = relaxation time

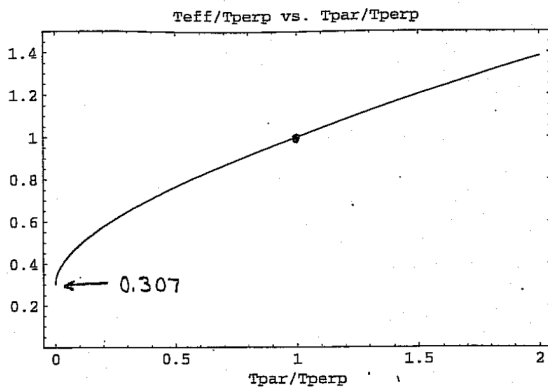
$$\begin{aligned} &= \frac{15(k_B \tilde{T}/mc^2)^{3/2} (4\pi\epsilon_0)^2 m^2 c^3}{8\pi^{1/2} q^4 \ln \Lambda n} \\ &= \left( \frac{15\pi^{1/2}}{8 \ln \Lambda} \right) \frac{1}{\nu_c}, \end{aligned}$$

where  $\nu_c$  is the collision frequency at thermal equilibrium:

$$\nu_c \sim \pi \left( \frac{q^2}{4\pi\epsilon_0 k_B T} \right)^2 n_0 \left( \frac{k_B T}{m} \right)^{1/2}.$$

1.5

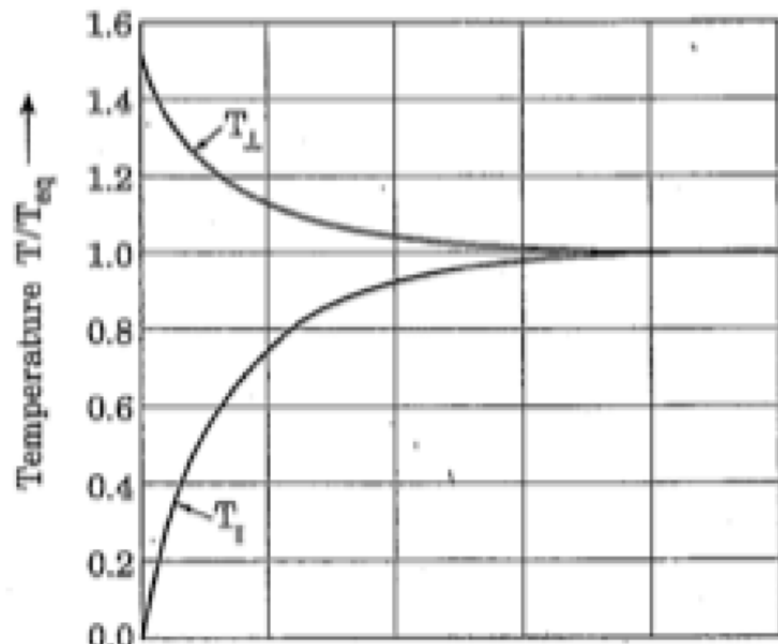
$$\frac{T_{\text{eff}}}{T_{\perp}}$$



[...]



1.6



## 1.7 The Boersch effect

Aren't collisions negligible? Putting in numbers for ions:

$$\tau_{\text{eff}} = 4.3 \cdot 10^{-4} [s] \left( \frac{A^{1/2}}{Z^4} \right) \left( \frac{k\tilde{T}}{[eV]} \right)^{3/2} \left( \frac{15}{\ln \Lambda} \right) \left( \frac{10^{10} [cm^{-3}]}{n} \right) \quad (1)$$

$$\Lambda = \frac{1.5 \cdot 10^5 (k_B T / [eV])^{3/2}}{Z^3 (n / 10^{10} [cm^{-3}])} \quad (2)$$

Example: 2 MeV injector:

[...]

# Outline

1. Coulomb collisions
2. Beam-gas scattering
3. Charge-changing processes
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability

## 2.1 Coulomb collisions in residual gas

Rate of change of momentum transverse to velocity:

$$\frac{dp_x}{dt} = \frac{ZZ_g e^2}{4\pi\epsilon_0 r^2} \frac{b}{r}$$

Total change in momentum:

$$\begin{aligned}\Delta p &= \int_{-\infty}^{\infty} \frac{dp_x}{dt} dt \\ &= \int_{-\infty}^{\infty} \frac{dp_x}{dt} \frac{dt}{dz} dz \\ &= \frac{ZZ_g e^2 b}{4\pi\epsilon_0 v} \int_{-\infty}^{\infty} \frac{dz}{(z^2 + b^2)^{3/2}} \\ &= \frac{2ZZ_g e^2 b}{4\pi\epsilon_0 v b}.\end{aligned}$$

Approximate deflection angle  $\theta$ :

$$\theta \approx \frac{\Delta p}{p} = \frac{2ZZ_g e^2 b}{4\pi\epsilon_0 p v b}$$

## 2.2 Multiple collisions

After traversing distance  $s$  and undergoing  $N_s$  collisions, the mean square angle  $\langle \Theta^2 \rangle$  can be written

$$\begin{aligned}\langle \Theta^2 \rangle &= N_s \langle \theta \rangle^2 \\ &= n_g \sigma_s s \langle \theta \rangle^2 \\ &= 8\pi n_g \left( \frac{ZZ_g e^2}{4\pi\epsilon_0 mc^2 \gamma \beta^2} \right)^2 \ln \left( \frac{\theta_{max}}{\theta_{min}} \right) s\end{aligned}$$

Writing

$$\langle \Theta^2 \rangle = \langle x'^2 \rangle + \langle y'^2 \rangle = 2\langle x'^2 \rangle,$$

and taking the  $s$ -derivative gives a dynamical equation for  $\langle x'^2 \rangle$ :

$$\frac{d}{ds} \langle x'^2 \rangle = 4\pi n_g \left( \frac{ZZ_g e^2}{4\pi\epsilon_0 mc^2 \gamma \beta^2} \right)^2 \ln \left( \frac{\theta_{max}}{\theta_{min}} \right) \quad (3)$$

$$\equiv C. \quad (4)$$

## 2.3 Jackson's approximation of $\theta_{min}$ and $\theta_{max}$

Jackson argues that  $\theta_{max}$  arises from the distributed nature of the nucleus and  $\theta_{min}$  from the screening of electrons or the quantum uncertainty principle.

$$\ln \left( \frac{\theta_{max}}{\theta_{min}} \right) \approx 2 \ln (204 Z_g^{-1/3})$$

## 2.4 How does scattering affect the envelope equations?

*Thin lens approximation:* assume the scattering locally changes the transverse momentum without directly changing the position.

Let the initial phase space coordinates be  $x = x_0$  and  $x' = x'_0$ . After an incremental distance  $\delta s$ ,

$$x \rightarrow x_0 \tag{5}$$

$$x' \rightarrow x'_0 + \delta x'. \tag{6}$$

We can then calculate the incremental change in the second-order moments:

$$\text{delta}\langle x'^2 \rangle = \langle \dots \rangle$$

# Outline

1. Coulomb collisions
2. Beam-gas scattering
- 3. Charge-changing processes**
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability



## 3.1

# Outline

1. Coulomb collisions
2. Beam-gas scattering
3. Charge-changing processes
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability



# Outline

1. Coulomb collisions
2. Beam-gas scattering
3. Charge-changing processes
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability



# Outline

1. Coulomb collisions
2. Beam-gas scattering
3. Charge-changing processes
4. Gas pressure instability
5. Electron cloud processes
6. Electron-ion instability



## 6.2 References

- [1] S. Ichimaru and M. N. Rosenbluth, *Relaxation Processes in Plasmas with Magnetic Field. Temperature Relaxations*, The Physics of Fluids **13**, 2778 (1970).